

1. GIT IS A DISTRIBUTED CONTROL VERSION SYSTEM, WHILE GITHUB IS A (A) CLOUD-BASED COLLABORATION PLATFORM.
- THE PRIMARY PURPOSE OF GIT IS TO BE THE AN INTERMEDIATE TO THE INFO COMMITTED TO GITHUB, PEOPLE CAN HAVE A PREVIEW OF THE DATA OR INFO BEFORE IT IS COMMITTED, ALSO IT HELPS WORK BEING EFFICIENT SINCE IT CAN BE USED OFFLINE AND ON ITS LOCAL REPOSITORIES AND EACH USER HAS A COMPLETE HISTORY VERSION OF IT. GITHUB ON THE OTHER HAND TAKES ALL THE INFO FROM GIT AND PUBLISHES IT, CREATING A COLLABORATIVE ENVIRONMENT IN WHICH INFO IS (B) REPOSITORY → LOCAL STORAGE IN WHICH THE COMPLETE HISTORY OF A PROJECT IS KEPT AT.
- COMMIT → SNAPSHOTS OF ALL THE CHANGES MADE ON A PROJECT TO KEEP TRACK OF THEM WHILE THEY ARE BEING "PUBLISHED", HELPS CHECK THE HISTORY OF IT.
- BRANCH → IS THE PARALLEL WORK OF A PROJECT OR A "COPY" OF CHANGES THAT HAVE BEEN WORKED ON A REPOSITORY WITHOUT INTERFERING ON THE ORIGINAL FILE. WHEN THE USER IS SURE OF THEM, HE WILL MERGE THEM OR REBASE THEM.
- STORED AT REMOTE REPOSITORIES. ALSO PEOPLE CAN TRACK ISSUES, ACTIONS, CF, CD, PULL REQUESTS AND CAN HAVE DOCUMENTATION LIKE WIKI INFO OR README FILES.

THE THREE MAIN CONCEPTS ARE IMPORTANT IN COLLABORATIVE ANALYTICS PROJECTS SINCE EVERY USER CAN WORK IN AN EFFICIENT WAY, WITHIN ALL THE HISTORY OF CHANGES OF A REPOSITORY OR FILE ALLOWING THEM TO REVIEW ALL OF THE CODE. IT IS ALSO VERY USEFUL FOR PEOPLE TO CHECK OUT CHANGES PRIOR TO THE PUBLICATION.

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2. a. PYTHON IS USEFUL FOR DATA ANALYSIS AND ANALYTICS, WHILE SQL ALSO HELPS WITH DATA BUT IN A ELT OR ETL WAY (DATA EXTRACTION, LOADING AND TRANSFORMATION (ELT) OR EXTRACTION, TRANSFORMATION AND LOADING (ETL)); SQL IS A LANGUAGE FORM SO IT MAY BE MORE SPECIALIZED. PYTHON CAN ALSO HELP RESHAPE DATA IN A SIMPLER FORM.

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- b. SYNTAX → THE WAY A CODE OR STRUCTURE IS WRITTEN HAS RULES FOR ITS CREATION IN ORDER FOR IT TO WORK PROPERLY IN A PROGRAMMING LANGUAGE.

LOOPS → THE WAY A PROGRAMMING LANGUAGE LEARNS TO BEHAVING AND REPEATS THE SAME ACTIONS.

LIBRARIES → GROUPS OR PACKAGES OF CODE WHICH ARE PRE-LOADED AND CAN HELP WITH SPECIFIC FUNCTIONS.

each one of them is important because they concatenate the entire way in which information can be created and modified, they depend on each other to have a correct way to handle and analyze data.

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c. Containers will have a big library of pre-loaded images, which can be accessed by from `hub.docker.com`, they are not heavy files, everyone can have access to them, and they have a big and varied catalogue of images.

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d. IDE $\rightarrow$ ~~the~~ integrated development environments, they are? programming languages that help ~~also~~ the development of other tools ~~rather~~ allowing them to work hand on hand, ~~like~~ like two popular IDE's: RSTUDIO R and VS CODE (VISUAL STUDIO CODE). RSTUDIO.

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3. n8n $\rightarrow$ is a low-code & tool which can create workflows
a. through the use of nodes.

QGIS $\rightarrow$ can help with geospatial and ~~geographic~~ geographic information since one can trace, through layers, all the components of a geographical space (e.g. Malawi). no-code.

KNIME $\rightarrow$ low-code tool which is ~~used~~ used for its customized and personalized visualization of information.

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[b] * THIS ANSWER IS AT THE END OF THE ANSWERS *

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4.a. mean: is the arithmetic average which can be used ~~for~~ when one wants distributed and ~~symmetric~~ symmetric participation of an item. it can be used when the ~~presence~~ presence of outliers doesn't mind, ~~like~~ like: the average number of seats a classroom can have.

median: the "real" average taking out the outliers, it gives a more precise result since it only takes into consideration the numbers that are closer to the center. it can be used to calculate the average age of a customer.

mode: the ~~mode~~ number of times a number repeats. as a ~~farther~~ farther. it can be used when a person wants to know how many times a value can appear, like when a nutritionist wants to know how many patients with the same ~~body~~ weight she has.

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b. univariate: no complex, uses and analyses one variable. e.g. a company wants to check the ~~total~~ monthly sales, ~~or something~~.

bivariate: complex, uses and compares two variables. e.g. a marketing team wants to see if the money spent on advertisements had an impact on sales.

MULTIVARIATE: more complex, uses and compares more than two variables at a time. e.g. a retail company wants to see if there is a relationship between sales, location, average ticket and seasonality in order for them to bet on new collections.

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c. ~~the~~ DUPLICATE RECORDS: information can be duplicated in different rows of a CSV or excel file, making final data not reliable. you can remove duplicates directly on R, in order for you to have the original info.

- MISSING VALUES: cells that have no info and can be counted as if they had. you can ask R to fill in with NA, so that they are not taken into consideration and a small mistake could bring big consequences.

- INCONSISTENT DATA: when words have typo mistakes like in names or places, you can ask R to take those examples and rename all the cells so that the info is now clean.

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5. a. $\text{result} (3 + \text{sqrt}(9) - 2^{\{2\}} * 4/3)$

b. `getwd()`

c. `?mean()`

d. `library(tidyverse)`
`region <- c("North", "South", "East")`
`sales <- c(45000, 38000, 52000)`
`sales_data <- tibble(region, sales))`

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3.b. non business the advantages of using no-code and low-code tools for analytics that can be mentioned are: you ~~don't~~ need the technical or coding ~~skills~~ "basic" skills to get an analysis, they can be customizable depending on the needs, most of them work in a similar way, you don't need to have a lot of excel sheets or documents, in which ~~more~~ mistakes can be easily made and

sometime) hard to select, analysis and visualization are friendly and with the use of AI, everything can be basically done. And once you have made all the changes, you can easily go back to adapt information or share it with others.

Some situations that can be named are: if you are an entrepreneur, if you want to deliver a ~~proper~~ professional and visual work on the results of a company presented to stakeholders in a "fast" way; or when you want to have your work saved on a cloud-based tool only feeding it with final and checked files to keep track of (for example) files.

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